Assignment

Q.1. Examine the convergence of the following series:

(a) $\left(\frac{2^2}{1^2}-\frac{2}{1}\right)^{-1}+\left(\frac{3^3}{2^3}-\frac{3}{2}\right)^{-2}+\left(\frac{4^4}{3^4}-\frac{4}{3}\right)^{-3}+\cdots\ldots\ldots\ldots$

(b) $\sum\left(\frac{n+1}{3n}\right)^{n}$

(c) $\frac{2}{1}-\frac{3}{2}+\frac{4}{3}-\frac{5}{4}+\cdots\ldots\ldots\ldots$

Q.2. If $\vec{r}=x\hat{\imath}+y\hat{\jmath}+z\hat{k}$, prove that $\vec{\nabla}\cdot(r^n\,\vec{r})=(n+3)\,r^n$.

Q.3. Show that $\int_C \vec{F}\cdot d\vec{r}=3\pi$, given that $\vec{F}=z\hat{\imath}+x\hat{\jmath}+y\hat{k}$ and C being the arc of the curve $\vec{r}=\cos t\ \hat{\imath}+\sin t\ \hat{\jmath}+t\,\hat{k}$ from $t=0$ to $t=2\pi$.

Q.6. If $u=x\phi\left(\frac{y}{x}\right)+\psi\left(\frac{y}{x}\right)$, prove that

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=0.$$

Q.7. Evaluate

(i) $\lim_{x\to 0}\frac{x^2+2\cos x-2}{x\sin^3 x}$ (ii) $\lim_{x\to 0}(\cos x)^{\cot^2 x}$

Q.8. Show that

(a) $\int_0^{\frac{\pi}{4}}\log(1+\tan\theta)\,d\theta=\frac{\pi}{8}\log 2$

(b) $\int_0^{\frac{\pi}{2}}\frac{\sin x\,dx}{\sin x+\cos x}=\frac{\pi}{4}$.

Q.9. Prove that

(a) $\int_0^{\infty}x^{2n-1}e^{-ax^2}=\frac{\Gamma(n)}{2a^n}$.

(b) $\int_0^1\left(\log\frac{1}{x}\right)^{n-1}dx=\Gamma(n)$.

Q.10. Show that

$$\int_0^1\frac{x^2\,dx}{(1-x^4)^{\frac{1}{2}}}\times\int_0^1\frac{dx}{(1+x^4)^{\frac{1}{2}}}=\frac{\pi}{4\sqrt{2}}.$$